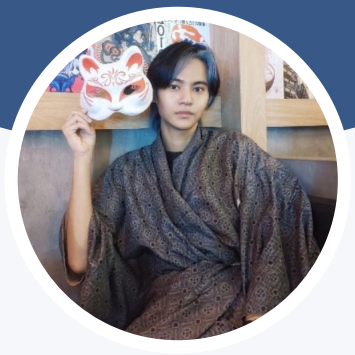
Kaung Sat Hein



Personal details

Kaung Sat Hein

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Yangon, Myanmar

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Skills

Languages & Frameworks
PHP (Laravel), Node.js, JavaScript, TypeScript, React.js

Databases & Caching
MySQL, MongoDB, Redis

Cloud & DevOps
AWS, Docker, CI/CD

Core Strengths
System Design, API Architecture, RBAC, Workflow Systems, Performance Optimization

Engineering Notes

Authorization Boundaries and RBAC Design
https://ksh-portfolio-nu.vercel.app/engineering-notes/authorization-boundaries-and-rbac-design

Profile

Backend-focused full stack developer specializing in **designing scalable, workflow-intensive business systems and high-performance backend architectures**.

Proven experience building and maintaining **HRMS, ATS, payroll, and internal platforms** supporting **100K+ users across multi-country environments**, with a strong focus on **data consistency, system reliability, and operational scalability**.

Skilled in **domain-driven design, RBAC systems, multi-level approval workflows, and complex business logic modeling**, enabling robust enterprise-grade applications. Strong background in **performance optimization (query design, indexing, caching), distributed system patterns, and API architecture**.
Experienced in integrating **AI-powered features, third-party services, and event-driven workflows**, delivering efficient, maintainable, and production-ready systems.

Employment

Web Backend Developer (Trainee) — **Oct 2023 - Jan 2024**
Better HR, Yangon

Web Backend Developer — **Feb 2024 - Present**
Better HR, Yangon

Selected Production Work

Better HR (HRMS Platform)
Implementation
Designed and implemented scalable backend systems including:

- Expense management & multi-level approvals
- KPI tracking & performance evaluation
- Permission groups (RBAC)
- Indonesia payroll system

Additional enhancements across HR workflows and system performance

Key Contributions

- Designed **multi-level approval workflows** with role-based access control
- Optimized database queries and implemented **Redis caching** for high-load systems
- Built **Excel import/export pipelines** for large datasets
- Improved system reliability for **100K+ users across 8 countries**

Impact

- Reduced manual HR operations and improved workflow efficiency
- Improved system performance and scalability under high concurrency through query optimization and caching strategies

Engineering Case Study — Problem → Solution → Trade-offs → Impact
https://ksh-portfolio-nu.vercel.app/projects/better-hr-hrms

System Design Tradeoffs
https://ksh-portfolio-nu.vercel.app/engineering-notes/system-design-tradeoffs

Database Indexing and Query Design
https://ksh-portfolio-nu.vercel.app/engineering-notes/database-indexing-and-query-design

Caching Strategies and Invalidation
https://ksh-portfolio-nu.vercel.app/engineering-notes/caching-strategies-and-invalidation

Audit Logs and Business Event Traceability
https://ksh-portfolio-nu.vercel.app/engineering-notes/audit-logs-and-business-event-traceability

Additional engineering notes
(system design, caching, RBAC)
https://ksh-portfolio-nu.vercel.app/engineering-notes

Courses

OpenAI API for Python Developers
Jan 2025

Explore a Career in SQL Development
Aug 2024

Become a Programmer: Foundations
Jul 2024

Docker Foundations Professional Certificate
May 2024

Career Essentials in Software Development by Microsoft and LinkedIn
May 2024

Introduction to Cloud Computing
Apr 2024

Build Your Laravel Skills
Apr 2024

Web Applications for Everybody Specialization
Mar 2023

Application Tracking System (ATS)
Implementation
Built AI-powered CV upload and parsing pipeline with structured extraction (CV → JSON) and atomic candidate creation in ATS.

Key Contributions
- Designed **state-machine-based workflow transitions**
- Built **AI-powered CV → structured profile extraction**
- Implemented **atomic upload-to-candidate pipeline**
- Integrated ATS with internal HR systems

Impact
- Streamlined hiring workflow and reduced manual processing
- Improved data consistency and candidate tracking

Engineering Case Study — Problem → Solution → Trade-offs → Impact
https://ksh-portfolio-nu.vercel.app/projects/application-tracking-system-ats

Job Content Management System (CMS)
Implementation
Developed a modular CMS system supporting job pages, companies, and search optimization.

Key Contributions
- Built **modular content sections & dynamic job pages**
- Designed **complex filtering & search logic**
- Integrated **Google Indexing API**
- Optimized CMS endpoints for performance

Impact
- Improved content management efficiency
- Enhanced search visibility and system scalability

Engineering Case Study — Problem → Solution → Trade-offs → Impact
https://ksh-portfolio-nu.vercel.app/projects/job-content-management-system-cms

Additional Systems
- **Middleware Service** — SMS/Email OTP system across Thailand & Vietnam
  https://ksh-portfolio-nu.vercel.app/projects/middleware-service
- **Customer Management System** — Multi-country lifecycle + Xero, Monday, Notion integration
  https://ksh-portfolio-nu.vercel.app/projects/customer-management-system-internal
- **HR AI Chatbot Agent** — Backend APIs for KPI, ATS, and performance data
  https://ksh-portfolio-nu.vercel.app/projects/hr-ai-chatbot-agent

Feature Highlights

ATS CV Upload & AI Profile Extraction
Designed async CV upload pipeline with file intake, AI extraction, profile creation, notifications, ATS synchronization; modeled structured applicant datasets.

System Design Deep Dive — Architecture & Implementation
https://ksh-portfolio-nu.vercel.app/projects/ats-cv-upload-ai-profile-extraction

Meta Front-End Developer
Jun 2023

Google IT Support
Sep 2023

HTML, CSS, and Javascript for Web Developers
Mar 2023

Expense Tracking System
Designed complete module scope including use cases, architecture, workflow, ERD, implementation plan, high-risk testing; modeled multi-level approval, paid-out handling, reclassification.

System Design Deep Dive — Architecture & Implementation
https://ksh-portfolio-nu.vercel.app/projects/expense-tracking-system

KPI Module
Designed full KPI module covering group management, target assignment, user tracking, exports, mobile visibility; structured setup, assignment, tracking, reporting.

System Design Deep Dive — Architecture & Implementation
https://ksh-portfolio-nu.vercel.app/projects/kpi-module

Permission Group
Designed group-based access control using department, position, location, status scope; separated permission definition, group mapping, runtime visibility evaluation.

System Design Deep Dive — Architecture & Implementation
https://ksh-portfolio-nu.vercel.app/projects/permission-group

Indonesia Payroll
Built payroll logic around annualized taxable income, personal/family relief, allowed deductions, progressive tax brackets, prior-tax reconciliation.

Payroll System Deep Dive — Tax Logic, Calculation & Reconciliation
https://ksh-portfolio-nu.vercel.app/projects/indonesia-payroll

Education

Bachelor of Engineering **Jan 2014 - Dec 2021**
West Yangon Technological University

Diploma, Computer Science **Apr 2024 - Apr 2025**
NCC Education